

## The Ferris Wheel

There are many rides at the amusement park whose movement can be described using trigonometric functions. The Ferris Wheel is a good example of periodic movement.

Sydney wants to ride a Ferris wheel that has a radius of 60 feet and is suspended 10 feet above the ground. The wheel rotates at a rate of 2 revolutions every 6 minutes. (Don't worry about the distance the seat is hanging from the bar.) Let the center of the wheel represents the origin of the axes.

- Write a function that describes a Sydney's height above the ground as a function of the number of seconds since she was  $\frac{1}{4}$  of the way around the circle (at the 3 o'clock position).
- How high is Sydney after 1.25 minutes?
- Sydney's friend got on after Sydney had been on the Ferris wheel long enough to move a quarter of the way around the circle. How would a graph of her friend's ride compare to the graph of Sydney's ride? What would the equation for Sydney's friend be?



## Water Wheel

Suppose that a water wheel with radius 7 inches is suspended 6 inches above the water. The water wheel rotates at five revolutions per minute (rpm). You start your stopwatch. Four seconds later, point  $P$  on the rim of the wheel is at its greatest height. You are to model the distance  $d$  of point  $P$  from the surface of the water in terms of the number of seconds  $t$ , the stopwatch reads.

- How long does it take point  $P$  to travel 1 revolution?
- Sketch one cycle of the graph and write an equation for a positive sine and positive cosine curve.
- What is the amplitude?
- What is the vertical shift?
- About how long does point  $P$  remain under water? (You may use your graphing calc)

